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BRUIN CLL-313: Randomized Phase III Trial of Pirtobrutinib Versus Bendamustine Plus Rituximab in Untreated Patients With Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

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ABSTRACT

PURPOSE BRUIN CLL-313 is a randomized, open-label, global phase III study comparing the efficacy and safety of pirtobrutinib, a highly selective, noncovalent Bruton tyrosine kinase inhibitor (BTKi), against bendamustine plus rituximab (BendaR), a common frontline chemoimmunotherapy, in treatment-naïve patients with chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL).

METHODS Patients with previously untreated CLL/SLL without del(17p) were randomly assigned 1:1 to continuous pirtobrutinib monotherapy or BendaR, stratified by immunoglobulin heavy chain gene mutation status and Rai stage. The primary end point was independent review committee (IRC)–assessed progression-free survival (PFS); secondary end points included overall survival (OS), investigator (INV)–assessed PFS, safety, and tolerability.

RESULTS Overall, 282 patients were randomly assigned to receive pirtobrutinib (n = 141) or BendaR (n = 141). IRC-assessed PFS was significantly improved with pirtobrutinib versus BendaR (hazard ratio [HR], 0.199 [95% CI, 0.107 to 0.367]; $P < .0001$), and the 24-month PFS rate was 93.4% (95% CI, 87.6 to 96.5) and 70.7% (95% CI, 61.5 to 78.1), respectively. INV-assessed PFS similarly favored pirtobrutinib (HR, 0.186 [95% CI, 0.093 to 0.371]). Interim analysis of OS favored pirtobrutinib (median follow-up 32 months; HR, 0.257 [95% CI, 0.070 to 0.934]) despite an effective crossover rate of 52.9%. In patients receiving pirtobrutinib versus BendaR: adverse event (AE)–related dose reductions occurred in 3.6% versus 31.1% of patients; grade ≥3 treatment-emergent AEs (TEAEs) occurred in 40.0% versus 67.4% of patients; and treatment discontinuations because of TEAEs occurred in 4.3% versus 15.2% of patients, respectively.

CONCLUSION Pirtobrutinib demonstrated superiority over BendaR in IRC-assessed PFS in treatment-naïve CLL/SLL. OS trends favored pirtobrutinib despite the study design allowing for crossover. Pirtobrutinib was well tolerated, consistent with its known safety profile, and more favorable than BendaR.

ACCOMPANYING CONTENT

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Appendix

Data Sharing Statement

Protocol

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INTRODUCTION

Treatment outcomes in patients with chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) have improved drastically in the past 10 years with Bruton tyrosine kinase inhibitors (BTKi) and B-cell lymphoma 2 inhibitors (BCL2i).[1] Recent phase III studies have established the covalent (c) BTKi ibrutinib, zanubrutinib, and acalabrutinib, and the BCL2i venetoclax combined with obinutuzumab, as preferred treatments over chemoimmunotherapy (CIT) in patients with treatment-naïve CLL/SLL.[2–5] However, the CIT regimen of bendamustine plus rituximab (BendaR) still continues to be globally used in patients receiving frontline therapy.[6,7]

Pirtobrutinib, a highly selective noncovalent (nc) BTKi, inhibits BTK throughout the once daily dosing interval with

CONTEXT

Key Objective

Here we report, to our knowledge, the first prospective randomized phase III clinical trial examining the efficacy and safety of a noncovalent Bruton tyrosine kinase inhibitor, pirtobrutinib, exclusively in patients with untreated chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL). The primary objective of BRUIN CLL-313 was to determine whether pirtobrutinib improved progression-free survival (PFS) compared to bendamustine plus rituximab (BendaR).

Knowledge Generated

In patients with treatment-naïve CLL/SLL, pirtobrutinib was well tolerated and demonstrated a statistically significant and clinically meaningful improvement in independent review committee–assessed PFS compared with BendaR.

Relevance *(J.W. Friedberg)*

This trial demonstrates the potential role of noncovalent BTK inhibitors as upfront therapy of CLL, although additional trials are required to define the optimal sequence of BTK inhibitors, and how pirtobrutinib would compare to doublet or triplet combinations. BendaR should no longer be used as a control arm for CLL trials.*

*Relevance section written by *JCO* Editor-in-Chief Jonathan W. Friedberg, MD.

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low nM potency.[8] In the first-in-human phase I/II BRUIN study, pirtobrutinib showed promising safety and efficacy in patients with relapsed/refractory CLL/SLL, including patients previously treated with a cBTKi, leading to accelerated approval by the US Food and Drug Administration (FDA) in patients who received at least two prior lines of therapy, including a BTKi and a BCL2i.[9,10] Furthermore, the randomized phase III BRUIN CLL-321 study established superiority of pirtobrutinib monotherapy compared with investigator (INV)'s choice of idelalisib plus rituximab or BendaR specifically in patients with CLL/SLL previously treated with a cBTKi,[11] with subsequent global approvals. Although regulatory approval of pirtobrutinib in the post-cBTKi setting is not yet endorsed by the FDA, pirtobrutinib is recommended by the National Comprehensive Cancer Network guideline panel in second-line or subsequent therapy after resistance or intolerance to prior cBTKi-based regimens for patients.[12]

Data from the first phase III trial specifically assessing the safety and efficacy of a ncBTKi, pirtobrutinib, versus BendaR in treatment-naïve patients with CLL/SLL are provided in this article.

METHODS

Patients

Eligible patients were adults age 18 years or older requiring treatment with a confirmed diagnosis of CLL/SLL per International Workshop on CLL (iwCLL) 2018 criteria[13] who were treatment-naïve without evidence of 17p deletion (del(17p)) and with an Eastern Cooperative Oncology Group (ECOG) score of 0-2. A priori testing for *TP53* mutations was not an eligibility requirement. Patients with controlled atrial fibrillation and those requiring concomitant anticoagulant (excluding any vitamin K antagonist) or antiplatelet agents at the time of enrollment were permitted. Patients with known or suspected Richter transformation or central nervous system involvement were excluded. Full inclusion/exclusion criteria can be found in the protocol.

Study Design and Treatment

LOXO-BTK-20023 (BRUIN CLL-313) is a phase III global, randomized, open-label study comparing pirtobrutinib as continuous monotherapy (Arm A) with BendaR as fixed-duration treatment of six cycles (Arm B) in patients with untreated CLL/SLL and without evidence of del(17p) (Fig 1). Enrolled patients were randomly assigned 1:1 to receive either pirtobrutinib monotherapy or BendaR. An interactive voice/web response system was used for random assignment, and patients were stratified by immunoglobulin heavy chain gene (IGHV) mutation status (mutated *v* unmutated) and Rai stage (low/intermediate: 0-II *v* high: III-IV). Patients randomly assigned to BendaR were given the opportunity to crossover to pirtobrutinib treatment if still fulfilling eligibility criteria including independent review committee (IRC)–confirmed progressive disease (PD) per iwCLL 2018 criteria.

Local results for del(17p) and IGHV mutation status were used for eligibility and stratification, respectively. Central testing results were used if local results were unavailable. Peripheral blood samples collected at baseline were tested centrally by fluorescence in situ hybridization (FISH) at Molecular Pathology Lab Network (MPLN, Maryville, TN), Q2 Solutions (Durham, NC) or KingMed (Guangzhou, Guangdong, CN), or by next-generation sequencing (NGS) at Foundation Medicine, Inc (FMI, Cambridge, MA), where allowed per country regulations.

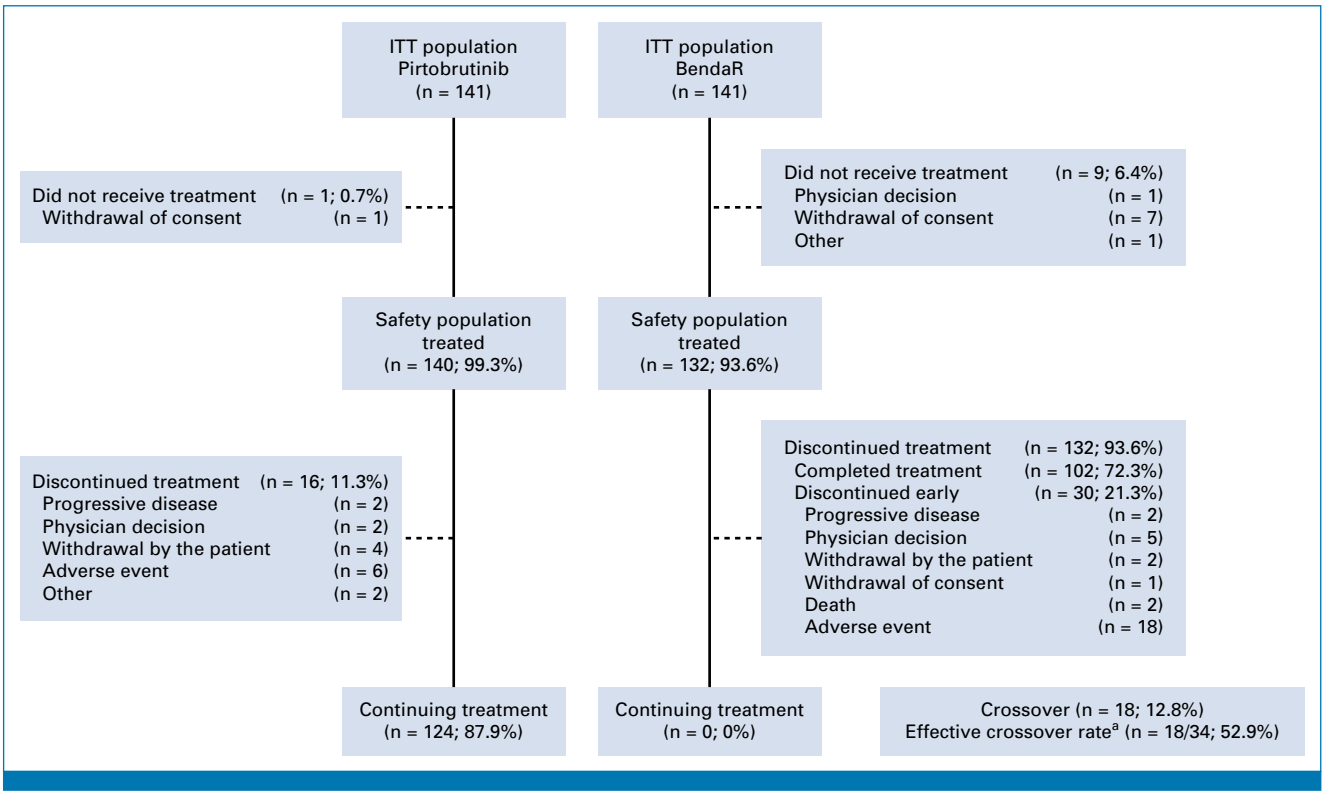

FIG 1. Flow diagram. [a]Effective crossover rate uses the number of participants with INV-assessed progressive disease as the denominator. INV, investigator; ITT, intention-to-treat.

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Deletion status of 17p and 11q was assessed by FISH, while IGHV and *TP53* mutation status were determined using NGS. Bone marrow aspirates and biopsies were evaluated centrally (MPLN or Kingmed).

Pirtobrutinib was administered continuously at 200 mg orally once daily. Patients randomly assigned to BendaR received bendamustine 90 mg/m² intravenously once daily on days 1 and 2 of each 28-day cycle, cycles 1 through six and rituximab 375 mg/m² once daily on day 1 for the first cycle and then 500 mg/m² once daily on day 1 of each 28-day cycle, cycles 2 through 6. Dose modifications were allowed for the management of study treatment-related toxicities.

The protocol was approved by the institutional review boards or independent ethics committees overseeing each site. The trial was conducted in compliance with the Declaration of Helsinki, Good Clinical Practice guidelines, and local laws. All patients provided written informed consent. The study was registered on ClinicalTrials.gov (identifier: NCT05023980)

Study Assessments

Response was assessed using iwCLL 2018 criteria.[13] The primary end point in this study was IRC-assessed progression-free survival (PFS), defined as the time from the date of random assignment until disease progression or death from any cause, whichever occurred first. Overall survival (OS) was a key secondary end point. Other secondary end points included INV-assessed PFS, overall response rate (ORR) assessed by both the INV and IRC, and safety measures. OS was defined as the time from random assignment until death from any cause and included the crossover period for patients in the BendaR arm who crossed over to receive pirtobrutinib. ORR was defined as the proportion of patients who achieved a best overall response of complete response (CR), CR with incomplete bone marrow recovery, nodular partial response (PR), and PR at or before the initiation of subsequent anticancer therapy.

Treatment-emergent adverse events (TEAEs) were defined as events that occurred or worsened in severity after baseline through 30 days (+7 days window) after the date of the last dose of study treatment or the first date of starting new anticancer therapy for CLL/SLL, whichever was earlier. AEs were graded by the INV according to the National Cancer Institute Common Terminology Criteria for Adverse Events v5.0 for nonhematologic and hematologic AEs. Treatment-related AEs are TEAEs considered related to study drug treatment by the INV.

Statistical Analysis

A sample size of 250 patients with 53 IRC-assessed PFS events was estimated to provide approximately 90% power to detect a hazard ratio (HR) of 0.41, corresponding to an estimated median IRC-assessed PFS of 43 months in the BendaR arm and 105 months in the pirtobrutinib arm, at a two-sided alpha

level of 0.05. OS was to be inferentially tested only if IRC-assessed PFS was significant. At the time of the final PFS analysis, an interim analysis of OS was performed with a small two-sided alpha of .000001 allocated.

The Kaplan-Meier method estimated PFS, OS, and other time-to-event endpoints, and differences between groups were evaluated using a stratified log-rank test. HR and 95% CI were estimated from stratified Cox proportional hazards models, except for subgroup analyses which utilized unstratified Cox proportional hazards models. Median follow-up for time-to-event end points were estimated using the reverse Kaplan-Meier method. ORR was provided with 95% CI and compared between groups using a stratified Cochran-Mantel-Haenszel test. All stratified analyses used the random assignment strata.

Due to differences in the duration of treatment between pirtobrutinib and BendaR (the latter being a fixed course), TEAEs were also summarized using exposure-adjusted incidence rates (EAIR) per 100 person-years. EAIR were based on the first occurrence of an event and were calculated as the number of events divided by the sum of years at risk for a TEAE across all patients times 100. EAIR ratios (EA-IRR) and 95% CI were estimated using Poisson regression and based on the EAIR with pirtobrutinib relative to the EAIR with BendaR.

Efficacy analyses were performed on the intention-to-treat (ITT) population, including all randomly assigned patients according to the treatment group they were assigned. Safety analyses included all randomly assigned patients who received at least one dose of study treatment and excluded the crossover period.

Statistical analyses were performed using SAS software (SAS 9.4, SAS Institute Inc, Cary, NC) and R software (version 4.4.2, R Core Team, Vienna, Austria). All *P* values are two-sided, and *P* values from tests of secondary end points that are not alpha-controlled are nominal. A data cutoff date of July 11, 2025, was used for all analyses.

RESULTS

Baseline Patient and Disease Characteristics

From December 6, 2021, to April 13, 2023, 282 patients with treatment-naïve CLL/SLL were randomly assigned to receive either pirtobrutinib (n = 141) or BendaR (n = 141; Fig 1). Overall, baseline patient and disease characteristics were similar between groups. The median age was 66 years (range, 29-88), 61% were male, and 94% had an ECOG PS 0+1 (Table 1). The prevalence of high-risk genomic features was generally balanced between the pirtobrutinib and BendaR groups, including unmutated IGHV (57% *v* 57%), and among patients with evaluable samples, complex karyotype (≥3 abnormalities; 16/91 [18%] *v* 11/83 [13%]), and *TP53* mutations (10/125 [8%] *v* 12/119 [10%]).

TABLE 1. Baseline Patient and Disease Characteristics

Characteristic	ITT Population	
	Pirtobrutinib (n = 141)	BendaR (n = 141)
Age, years		
Median (range)	65 (29-84)	66 (35-88)
≥65, No. (%)	73 (51.8)	84 (59.6)
≥75, No. (%)	20 (14.2)	26 (18.4)
Sex, No. (%)		
Male	83 (58.9)	90 (63.8)
Female	58 (41.1)	51 (36.2)
Region, No. (%)		
North America	2 (1.4)	5 (3.5)
Europe	85 (60.3)	85 (60.3)
Asia	32 (22.7)	31 (22.0)
South America	9 (6.4)	13 (9.2)
Australia/New Zealand	13 (9.2)	7 (5.0)
Histology, No. (%)		
CLL	131 (92.9)	129 (91.5)
SLL	10 (7.1)	12 (8.5)
Rai stage,[a] No. (%)		
0-II	82 (62.6)	81 (62.8)
III/IV	49 (37.4)	48 (37.2)
Baseline ECOG PS, No. (%)		
0	68 (48.2)	55 (39.0)
1	66 (46.8)	77 (54.6)
2	7 (5.0)	9 (6.4)
Duration of disease, years		
Median (Q1, Q3)	2.5 (0.8, 4.9)	2.0 (0.3, 4.7)
Bulky disease, No. (%)		
≥5 cm	39 (27.7)	35 (24.8)
β2 microglobulin, No. (%)		
≤3.5 mg/L	67 (47.5)	64 (45.4)
>3.5 mg/L	73 (51.8)	75 (53.2)
Missing	1 (0.7)	2 (1.4)
IGHV mutation status,[b] No. (%)		
Mutated	61 (43.3)	61 (43.3)
Unmutated	80 (56.7)	80 (56.7)
TP53 mutation status[c]		
No./n available (%)	10/125 (8.0)	12/119 (10.1)
Complex karyotype[c,d]		
No./n available (%)	16/91 (17.6)	11/83 (13.3)

Abbreviations: BendaR, bendamustine plus rituximab; CLL, chronic lymphocytic leukemia; ECOG PS, Eastern Cooperative Oncology Group performance status; IGHV, immunoglobulin heavy chain gene; ITT, intention-to-treat; SLL, small lymphocytic lymphoma.

[a]Rai stage as reported via electronic data capture, with counts and percentages based on the number of participants with CLL (not SLL); 131 in Arm A and 129 in Arm B.

[b]Local or central laboratory test as reported via electronic data capture.

[c]Central laboratory test.

[d]≥3 abnormalities in the same clonal population.

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At the time of data cutoff, 124 of 141 (87.9%) patients remained on pirtobrutinib treatment, and all patients randomly assigned to BendaR were off treatment, with 102 (72.3%) patients completing treatment. One patient assigned to pirtobrutinib and nine patients assigned to BendaR did not receive any treatment. Of 34 patients on BendaR who experienced PD as assessed by INV, 18 crossed over after fulfilling initial eligibility criteria and IRC-confirmed PD, for an effective crossover rate of 52.9% (n = 18/34).

Efficacy

Pirtobrutinib demonstrated superiority in the primary end point of IRC-assessed PFS compared with BendaR (HR, 0.199 [95% CI, 0.107 to 0.367]; $P < .0001$), with the risk of progression or death reduced by 80.1%. The median follow-up for IRC-assessed PFS was 28.1 months. The median PFS by IRC with pirtobrutinib was not reached, and the 24-month PFS rate was 93.4% (95% CI, 87.6 to 96.5), whereas the median PFS with BendaR was 33.5 months (95% CI, 32.7 to not estimable [NE]) and the 24-month PFS rate was 70.7% (95% CI, 61.5 to 78.1; Fig 2A). The difference in 24-month PFS rates was 22.7% (95% CI, 13.4 to 32.0). The PFS by INV assessment was consistent with these results (HR, 0.186 [95% CI, 0.093 to 0.371]; $P < .0001$; Appendix Fig A1, online only). IRC-PFS benefit was consistently observed with pirtobrutinib among prespecified, clinically relevant patient subgroups, including those age ≥65 years (HR, 0.24 [95% CI, 0.10 to 0.54]), those age ≥75 years (HR, 0.34 [95% CI, 0.07 to 1.69]), and in patients with high Rai stage (HR, 0.35 [95% CI, 0.14 to 0.88]), *TP53* mutations (HR, 0.14 [95% CI, 0.02 to 1.19]), complex karyotype (HR, 0.16 [95% CI, 0.02 to 1.41]), mutated IGHV (HR, 0.29 [95% CI, 0.09 to 0.91]), and unmutated IGHV (HR, 0.17 [95% CI, 0.08 to 0.36]; Fig 2B and Appendix Fig A2). The 24-month IRC-assessed PFS rates in patients with mutated IGHV were 96.7% (95% CI, 87.3 to 99.2) and 86.1% (95% CI, 73.1 to 93.1) in pirtobrutinib and BendaR groups, respectively, while the 24-month IRC-assessed PFS rates in patients with unmutated IGHV were 90.8% (95% CI, 81.6 to 95.5) and 58.6% (95% CI, 45.7 to 69.4), respectively.

ORR assessed by IRC was 94.3% (133/141, 95% CI, 89.1 to 97.5) in the pirtobrutinib group compared with 80.9% (114/141, 95% CI, 73.4 to 87.0) in the BendaR group ($P = .0006$). The median time to best response (PR or better) as assessed by the IRC was 3.9 months in the pirtobrutinib group and 4.0 months in the BendaR group. With pirtobrutinib monotherapy, 19 (13.5%) patients had a CR and 114 (80.9%) patients had a PR, and with BendaR, 29 (20.6%) patients had a CR and 85 (60.3%) patients had a PR (Table 2). INV-assessed ORR was consistent with these results (Table 2).

At a median follow-up for OS of 32.2 months, 3 (2.1%) deaths occurred in the pirtobrutinib group and 10 (7.1%) deaths in the BendaR group (Appendix Table A1). OS trended in favor of pirtobrutinib (HR, 0.257 [95% CI, 0.070 to 0.934]; $P = .0261$) at this interim analysis for OS (two-sided alpha = .000001). Final testing of OS superiority is planned at the end of the study. The median OS for patients in both arms was not reached (95% CI, NE to NE), and the 24-month OS rates were 97.8% (95% CI, 93.3 to 99.3) and 93.0% (95% CI, 87.0 to 96.3) for pirtobrutinib and BendaR, respectively (Fig 3). Of note, these results were observed despite an effective crossover rate of 52.9%. The results were consistent in a sensitivity analysis censoring 18 patients at the time of crossover in the BendaR arm (HR, 0.249 [95% CI, 0.068 to 0.906]).

Safety

Among 140 patients treated with pirtobrutinib and 132 patients treated with BendaR in the safety population, the median time on pirtobrutinib treatment was 32.3 months, and the median time on BendaR treatment was 5.6 months.

TEAEs of any grade occurred in 131 (93.6%) patients receiving pirtobrutinib and 117 (88.6%) patients receiving BendaR. COVID-19 disease was the most frequently occurring TEAE in patients receiving pirtobrutinib (21.4%), followed by upper respiratory tract infection (17.9%), arthralgia (12.1%), and neutropenia (12.1%). The most frequently occurring TEAEs in patients receiving BendaR were neutropenia (38.6%), nausea (23.5%), and pyrexia (18.9%). COVID-19 disease occurred in 9.1% of patients receiving BendaR (Appendix Table A2). Notably, TEAE of any grade hypertension occurred in <10% of patients, 7.9% with pirtobrutinib and 4.5% with BendaR.

The incidence of grade ≥3 TEAEs was 40.0% with pirtobrutinib and 67.4% with BendaR. The most common grade ≥3 TEAE in both arms was neutropenia (pirtobrutinib 7.1% *v* BendaR 34.8%). The number of patients who received growth factor support was 15 (10.7%) in the pirtobrutinib group and 81 (61.4%) in the BendaR group. Four (3.0%) patients receiving BendaR experienced grade ≥3 infusion-related reactions. Grade ≥3 TEAE of hypertension occurred infrequently, in 2.9% of patients receiving pirtobrutinib and 3.0% of patients receiving BendaR.

Since the median treatment duration was notably longer with pirtobrutinib than with BendaR, exposure-adjusted safety analyses were performed to account for these differences (Table 3). With the exception of arthralgia, in which the EAIR was higher with pirtobrutinib than with BendaR (EA-IRR, 3.05 [95% CI, 0.41 to 22.89]), the EAIR for all other common TEAEs was lower with pirtobrutinib than with BendaR.

AEs of special interest are summarized in Appendix Table A2 and Table 3. Infections were more frequent with pirtobrutinib (any grade 57.1%, grade ≥3 13.6%) than with BendaR (any grade 33.3%; grade ≥3 8.3%), but this may have been related to longer time on therapy. The EA-IRRs of pirtobrutinib versus BendaR were 0.43 (95% CI, 0.30 to 0.62) for any-grade infection, and 0.29 (95% CI, 0.14 to 0.61) for grade ≥3 infections. Any-grade bleeding was higher with pirtobrutinib (25.7%) than with BendaR (1.5%), though only one patient receiving pirtobrutinib experienced a grade ≥3 hemorrhage. The rates of atrial fibrillation and atrial flutter

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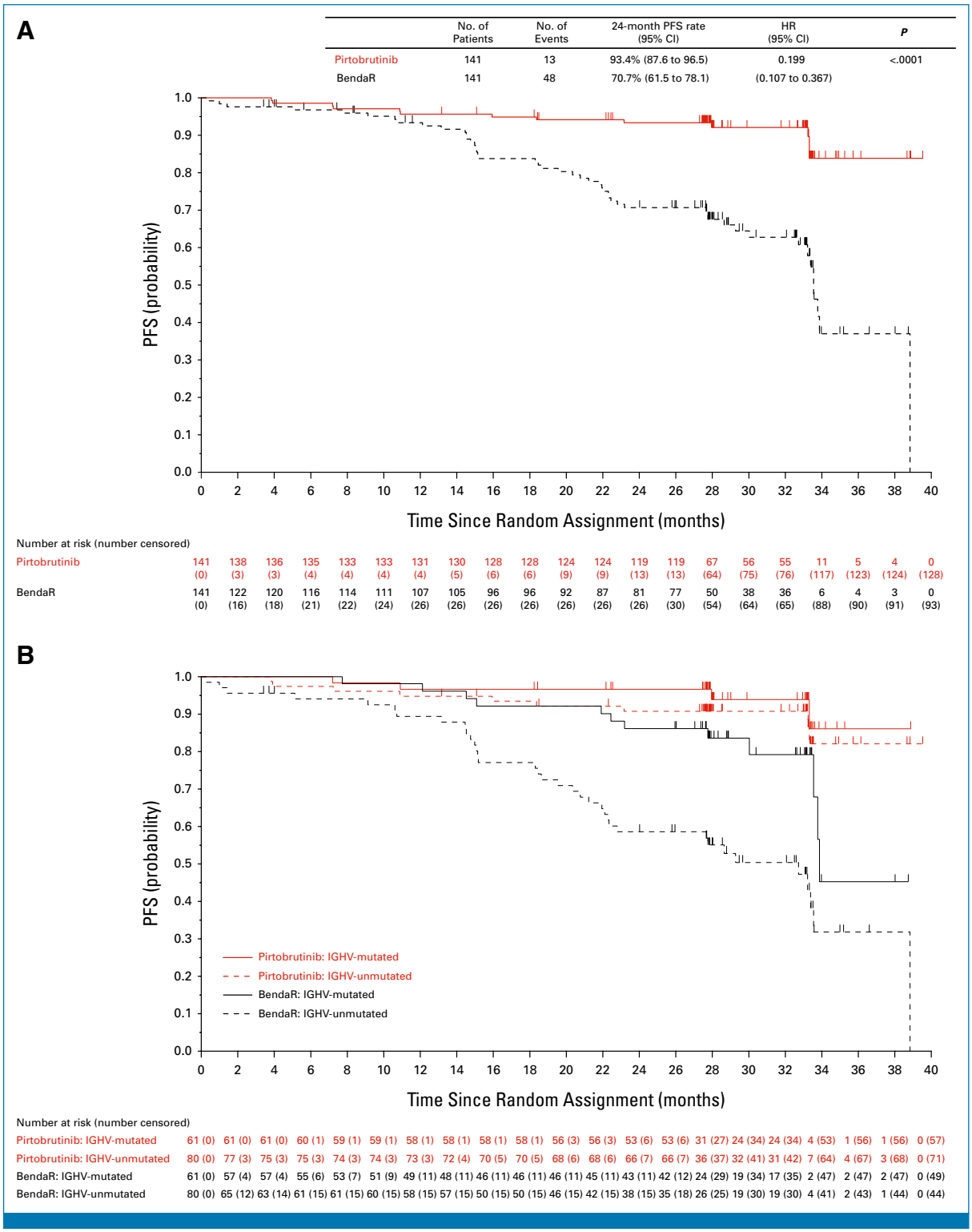

FIG 2. Kaplan-Meier curve of (A) IRC-assessed PFS and (B) IRC-assessed PFS by IGHV mutation status. BendaR, bendamustine plus rituximab; HR, hazard ratio; IGHV, immunoglobulin heavy chain gene; INV, Investigator; IRC, independent review committee; PFS, progression-free ssurvival.

were low across both pirtobrutinib (any grade 1.4%; grade ≥3 0.7%) and BendaR (any grade 1.5%; grade ≥3 0.8%).

AE-related dose reductions occurred in five (3.6%) patients receiving pirtobrutinib and 41 (31.1%) patients receiving BendaR. Treatment discontinuations because of TEAE, including deaths due to TEAE, occurred in six (4.3%) patients receiving pirtobrutinib (three [2.1%] considered treatment related) and 20 (15.2%) patients receiving BendaR (17 [12.9%] considered treatment related; Appendix Table A3).

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TABLE 2. Overall Response Rate and Best Overall Response as Assessed by IRC and Investigator

End Point	IRC-Assessed ORR		INV-Assessed ORR	
	Pirtobrutinib (n = 141)	BendaR (n = 141)	Pirtobrutinib (n = 141)	BendaR (n = 141)
ORR (PR or better)				
No. (%)	133 (94.3)	114 (80.9)	133 (94.3)	116 (82.3)
95% CI	89.13 to 97.52	73.38 to 86.99	89.13 to 97.52	74.95 to 88.18
P[a]	.0006		.0017	
Best overall response, No. (%)				
CR	19 (13.5)	29 (20.6)	8 (5.7)	29 (20.6)
CRi	0	0	0	0
nPR	0	0	12 (8.5)	1 (0.7)
PR	114 (80.9)	85 (60.3)	113 (80.1)	86 (61.0)
PR-L	0	0	2 (1.4)	0
SD	3 (2.1)	10 (7.1)	1 (0.7)	6 (4.3)
PD	2 (1.4)	1 (0.7)	2 (1.4)	3 (2.1)
UNK/NA	3 (2.1)	16 (11.3)	3 (2.1)	16 (11.3)

Abbreviations: BendaR, bendamustine plus rituximab; CR, complete response; CRi, CR with incomplete bone marrow recovery; INV, Investigator; IRC, Independent Review Committee; NA, not applicable; nPR, nodular partial response; ORR, overall response rate; PD, progressive disease; PR, partial response; PR-L, PR with lymphocytosis; SD, stable disease; UNK, unknown.
[a]Nominal two-sided *P* value.

No deaths due to COVID-19 disease were reported for either arm. Grade 5 TEAEs occurred in one (0.7%; septic shock) patient with pirtobrutinib and four (3.0%; pneumonia, cardiorespiratory arrest, ischemic stroke, tumor lysis syndrome) patients with BendaR, with none considered related to pirtobrutinib and one (tumor lysis syndrome) considered

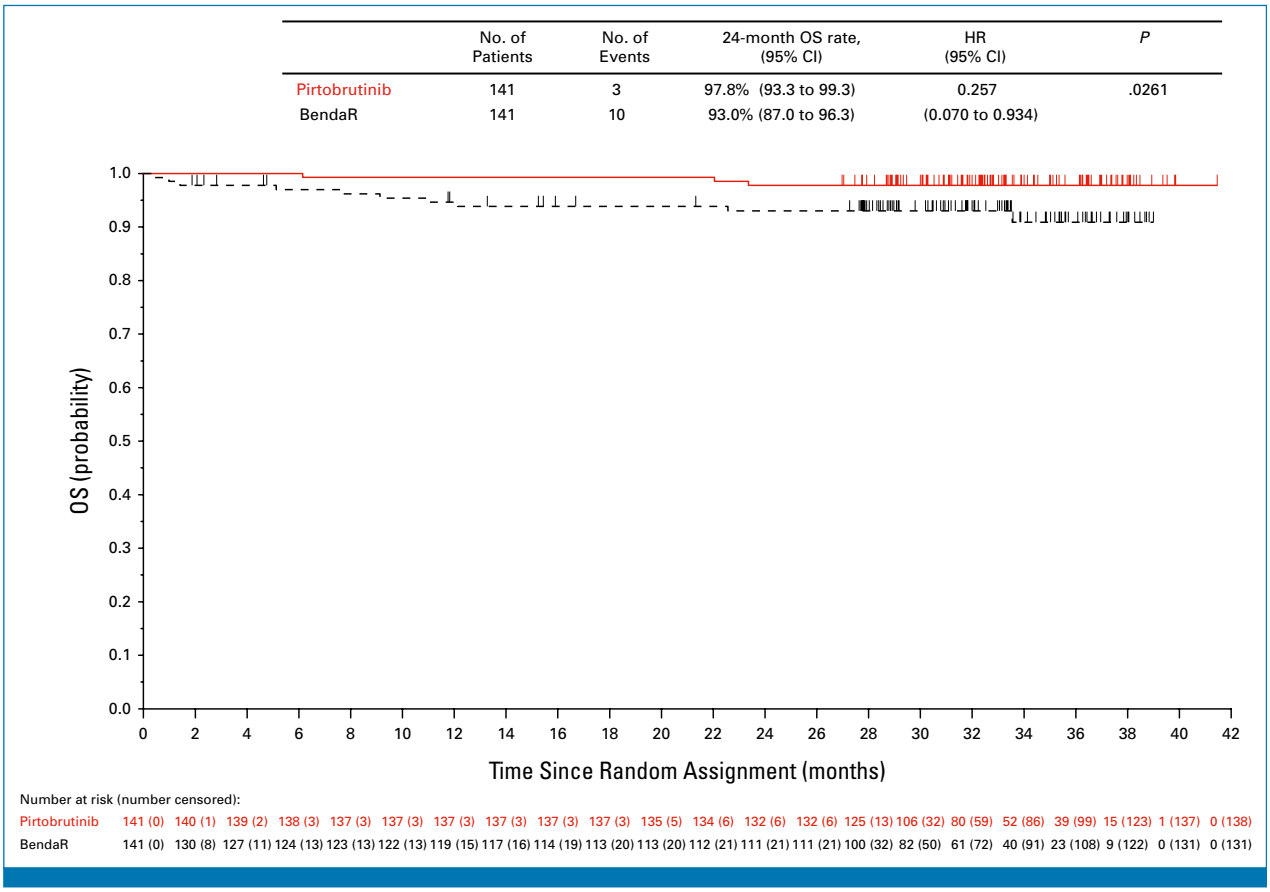

FIG 3. Kaplan-Meier curve of OS. BendaR, bendamustine plus rituximab; HR, hazard ratio; OS, overall survival.

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TABLE 3. Exposure-Adjusted Incidence Rates of TEAEs Occurring in ≥10% of Patients in Either Arm, and Exposure-Adjusted Incidence Rates for TEAEs of Special Interest

TEAE	Pirtobrutinib (n = 140), EAIR[a]	BendaR (n = 132), EAIR[a]	EA-IRR (95% CI)[b]
COVID-19	9.9	20.9	0.47 (0.24 to 0.92)
Upper respiratory tract infection	7.7	15.7	0.49 (0.23 to 1.05)
Arthralgia	5.1	1.7	3.05 (0.41 to 22.89)
Neutropenia	5.2	110.0	0.05 (0.03 to 0.08)
Back pain	4.4	8.5	0.52 (0.19 to 1.42)
Rash	4.4	34.0	0.13 (0.07 to 0.26)
Anemia	3.8	37.7	0.10 (0.05 to 0.20)
Pyrexia	3.5	49.1	0.07 (0.04 to 0.14)
Thrombocytopenia	3.2	30.7	0.10 (0.05 to 0.22)
Vomiting	2.8	27.4	0.10 (0.05 to 0.23)
Nausea	0.8	65.1	0.01 (0.00 to 0.04)
Neutrophil count decreased	0.5	25.1	0.02 (0.00 to 0.10)
Infusion-related reaction	0	39.0	–
AESI			
Infection[c]	38.3	89.7	0.43 (0.30 to 0.62)
Infection without COVID-19	30.9	74.9	0.41 (0.28 to 0.61)
Bleeding	12.5	3.3	3.73 (0.90 to 15.50)
Hemorrhage	5.2	3.3	1.55 (0.36 to 6.69)
Bruising	4.8	0	–
Petechiae and purpura	2.3	0	–
Neutropenia[d]	6.5	169.5	0.04 (0.02 to 0.06)
Anemia[e]	4.1	37.7	0.11 (0.06 to 0.21)
Thrombocytopenia[f]	3.5	43.1	0.08 (0.04 to 0.16)
Atrial fibrillation and atrial flutter	0.5	3.3	0.17 (0.02 to 1.17)

Abbreviations: BendaR, bendamustine plus rituximab; EA-IRR, EAIR ratios; EAIR, exposure-adjusted incidence rates; TEAE, treatment emergent adverse event.

[a]EAIR per 100 person-years is based on the first occurrence of an event and is calculated as the number of events divided by the sum of years at risk of a TEAE across all patients times 100.

[b]EA-IRR and 95% CI were estimated using Poisson regression and based on the EAIR with pirtobrutinib relative to the EAIR with BendaR.

[c]Aggregate of all preferred terms indicating infection and including COVID-19 disease. For grade ≥3 infections, the pirtobrutinib EAIR per 100 person-years was 5.5 and the BendaR EAIR per 100 person-years was 19.1, with EA-IRR of 0.29 (95% CI, 0.14 to 0.61).

[d]Includes neutropenia, neutrophil count decreased, and febrile neutropenia.

[e]Includes anemia and iron deficiency anemia.

[f]Includes thrombocytopenia and platelet count decreased.

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related to BendaR treatment. The case of tumor lysis syndrome began 4 days after treatment initiation and death occurred the following day for this patient.

DISCUSSION

In conclusion, to our knowledge, BRUIN CLL-313 is the first prospective, randomized phase III study examining the efficacy and safety of a ncBTKi, pirtobrutinib, exclusively in patients with untreated CLL/SLL.

Pirtobrutinib demonstrated a statistically significant and clinically meaningful improvement in IRC-assessed PFS compared with BendaR, and the primary end point was met. With a median follow-up of 28.1 months, patients in the pirtobrutinib group had significantly longer IRC-assessed PFS than patients in the BendaR group, and the risk of progression or death was reduced by 80.1%. Pirtobrutinib treatment benefit was also observed across clinically relevant subgroups, including those with high-risk molecular features such as *TP53* mutations, complex karyotype, and unmutated IGHV (Appendix Fig A2). Improvements in efficacy were also consistently observed among INV assessments.

The magnitude of the treatment effect for IRC-assessed PFS, favoring pirtobrutinib over BendaR, was large (HR, 0.199 [95% CI, 0.107 to 0.367]). In studies comparing ibrutinib (Alliance A041202)[14] and zanubrutinib (SEQUOIA)[3] with BendaR, the PFS HR was 0.39 (95% CI, 0.26 to 0.58) and 0.42 (95% CI, 0.28 to 0.63), respectively. Elevate TN also reported a low PFS HR, although with a different control arm (chlorambucil plus obinutuzumab) and eligibility criteria.[4] Although approximately 48% of patients in the pirtobrutinib group were <65 years, the benefit of pirtobrutinib also

extended to patients age ≥65 years in this study (HR, 0.24 [95% CI, 0.10 to 0.54]) and thus it does not appear that the treatment effect was driven by a favorable treatment effect in a younger patient population. Furthermore, in patients with mutated IGHV where CIT regimens appear particularly effective, pirtobrutinib demonstrated a benefit in PFS compared with BendaR (HR, 0.29 [95% CI, 0.09 to 0.91]). Zanubrutinib also showed some improvement in PFS versus BendaR in patients with mutated IGHV (HR, 0.67 [95% CI, 0.36 to 1.22]).[3]

Although OS was not yet mature at this analysis, OS favored pirtobrutinib (HR, 0.257 [95% CI, 0.070 to 0.934]), despite an effective crossover rate of 52.9%. This study continues to follow patients for OS.

The safety profile of patients receiving pirtobrutinib in this study was similar to that reported in previous studies with pirtobrutinib in CLL/SLL.[11,15] In this study, a more favorable safety profile was observed with pirtobrutinib than with BendaR, a difference that was more pronounced when accounting for duration of drug exposure, with a low frequency of pirtobrutinib-related discontinuations. Arthralgia had been previously reported in prior studies with pirtobrutinib, keeping with the same pattern, with the great majority of events considered to be low grade, with only a single patient reported as experiencing a grade ≥3 TEAE. Consistent with the known toxicities associated with BTKi,[16] bleeding AEs occurred more frequently in patients treated with pirtobrutinib (25.7%) than with BendaR (1.5%), but the majority of these were low-grade and grade ≥3 events were rare (hemorrhage 0.7%). Consistent with previous studies of pirtobrutinib,[9,11] rates of atrial fibrillation and atrial flutter were low in the pirtobrutinib group, similar to BendaR which has no known increased risk of these AEs. This was also observed with zanubrutinib in SEQUOIA,[3] but rates of any-grade atrial fibrillation were considerably higher with ibrutinib (17%) compared with BendaR (3%) in Alliance A041202.[14] In this study, the increased frequency of infections with pirtobrutinib compared with BendaR may have been related to the longer time on therapy.

These data combined with BRUIN-CLL 314 (ClinicalTrials.gov identifier: NCT05254743) raise important questions on optimal choice of BTKi therapy in frontline CLL. Factors such as patient age, comorbidities, and genetic risk factors are important to consider while making first-line (1L) treatment choice. Future research should include examining outcomes of cBTKi after ncBTKi, and additional combination regimens to address sequencing questions. While direct comparisons against current standard of care 1L treatments (continuous or time-limited) remain an important area of research, pirtobrutinib is also being investigated as part of combination therapy in ongoing phase III studies in both the treatment-naïve and R/R setting.[17,18]

This study has some important limitations. First, it excluded patients with del(17p) as these patients were not considered appropriate for the control arm. Thus, these data cannot be generalized to newly diagnosed patients with del(17p), although this patient population was included in the phase III study comparing pirtobrutinib with ibrutinib (ClinicalTrials.gov identifier: NCT05254743) that recently announced meeting its primary end point and is currently being recruited to single-arm cohorts of pirtobrutinib across two studies (ClinicalTrials.gov identifiers: NCT05254743, NCT06588478). Additionally, in the relapsed/refractory setting, it was shown that patients with del(17p) received benefit from treatment with pirtobrutinib.[9,11] As in other trials, this study was open label and thus potentially subject to INV bias. However, the primary end point of PFS was determined by a blinded IRC to minimize the impact of the open-label design, and data were also blinded to the study team until the primary analysis. Additionally, a median follow-up of 28.1 months for PFS in the frontline setting for CLL/SLL may be considered relatively short and may have contributed to a less reliable estimate for median PFS by IRC in the BendaR group. The 24-month PFS rates, however, which may be considered more stable given the follow-up, appear similar to those reported in other comparable studies.[3,14] Finally, OS remained immature with follow-up ongoing. An analysis with longer follow-up is planned to further understand the performance of pirtobrutinib in treatment-naïve CLL/SLL.

In summary, this study demonstrates the superiority of pirtobrutinib monotherapy compared with BendaR, as demonstrated by a statistically significant and clinically meaningful improvement in IRC-assessed PFS in patients with treatment-naïve CLL/SLL. This is now the third positive phase III study of pirtobrutinib in CLL/SLL, demonstrating the breadth of activity of pirtobrutinib across all lines of therapy in this disease.

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Eli Lilly provides access to all individual participant data collected during the trial, after anonymization, with the exception of pharmacokinetic or genetic data. Data are available to request 6 months after the indication studied has been approved in the United States and the European Union and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data-sharing environment. For details on submitting a request, see the instructions provided at www.vivli.org.

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APPENDIX

TABLE A1. Summary of Deaths of ITT Population

Death	Pirtobrutinib (n = 141)	BendaR (n = 141)
No. of patient deaths, No. (%)	3 (2.1)	10 (7.1)
Reason for death, No. (%)		
AE	1 (0.7)	6 (4.3)
Septic shock	1 (0.7)	0
Pneumonia	0	1 (0.7)
Cardiorespiratory arrest	0	1 (0.7)
Ischemic stroke	0	1 (0.7)
Tumor lysis syndrome	0	1 (0.7)
Hepatocellular carcinoma	0	1 (0.7)
Acute respiratory failure[a]	0	1 (0.7)
Disease progression	0	1 (0.7)
Other	2 (1.4)	3 (2.1)
Other[b]	1 (0.7)	2 (1.4)
Unknown	1 (0.7)	1 (0.7)

Abbreviations: AE, adverse event; BendaR, bendamustine plus rituximab; ITT, intention-to-treat.
[a]This patient did not receive treatment with BendaR.
[b]In the pirtobrutinib arm: patient did not return to site and cause of death was not determined. In the BendaR arm, both patients died during long-term follow-up: one patient died due to liver failure, and one patient died due to pneumonia.

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TABLE A2. TEAEs Occurring in ≥10% of Patients in Either Treatment Group and TEAEs of Special Interest Occurring in Either Treatment Group

	Pirtobrutinib (n = 140)		BendaR (n = 132)	
Preferred Term	Any Grade, No. (%)	Grade ≥3, No. (%)	Any Grade, No. (%)	Grade ≥3, No. (%)
Patients with ≥1 TEAE	131 (93.6)	56 (40.0)	117 (88.6)	89 (67.4)
COVID-19	30 (21.4)	1 (0.7)	12 (9.1)	2 (1.5)
Upper respiratory tract infection	25 (17.9)	1 (0.7)	9 (6.8)	0
Arthralgia	17 (12.1)	1 (0.7)	1 (0.8)	0
Neutropenia	17 (12.1)	10 (7.1)	51 (38.6)	46 (34.8)
Back pain	15 (10.7)	1 (0.7)	5 (3.8)	0
Rash	15 (10.7)	0	18 (13.6)	2 (1.5)
Anemia	13 (9.3)	6 (4.3)	21 (15.9)	10 (7.6)
Pyrexia	12 (8.6)	0	25 (18.9)	0
Thrombocytopenia	11 (7.9)	4 (2.9)	17 (12.9)	7 (5.3)
Vomiting	10 (7.1)	0	15 (11.4)	0
Nausea	3 (2.1)	0 (0)	31 (23.5)	1 (0.8)
Neutrophil count decreased	2 (1.4)	1 (0.7)	14 (10.6)	10 (7.6)
Infusion-related reaction	0	0	20 (15.2)	4 (3.0)
AE of special interest				
Patients with ≥1 AESI	103 (73.6)	34 (24.3)	93 (70.5)	69 (52.3)
Infection[a]	80 (57.1)	19 (13.6)	44 (33.3)	11 (8.3)
Infection without COVID-19	72 (51.4)	19 (13.6)	38 (28.8)	9 (6.8)
Bleeding	36 (25.7)	1 (0.7)	2 (1.5)	0
Hemorrhage	17 (12.1)	1 (0.7)	2 (1.5)	0
Bruising	16 (11.4)	0	0	0
Petechiae and purpura	8 (5.7)	0	0	0
Neutropenia[b]	21 (15.0)	13 (9.3)	68 (51.5)	60 (45.5)
Anemia[c]	14 (10.0)	6 (4.3)	21 (15.9)	10 (7.6)
Thrombocytopenia[d]	12 (8.6)	4 (2.9)	23 (17.4)	9 (6.8)
Atrial fibrillation and atrial flutter	2 (1.4)	1 (0.7)	2 (1.5)	1 (0.8)

Abbreviations: AE, adverse event; AESI, AE of Special Interest; BendaR, bendamustine plus rituximab; TEAE, treatment-emergent adverse event.
[a]Aggregate of all preferred terms indicating infection and including COVID-19 disease.
[b]Includes neutropenia, neutrophil count decreased, and febrile neutropenia.
[c]Includes anemia and iron deficiency anemia.
[d]Includes thrombocytopenia and platelet count decreased.

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TABLE A3. TEAEs Leading to Treatment Discontinuation

AE	Pirtobrutinib (n = 140)		BendaR (n = 132)	
	Any Grade, No. (%)	Grade ≥3, No. (%)	Any Grade, No. (%)	Grade ≥3, No. (%)
No. of patients with AE leading to discontinuation	6 (4.3)	3 (2.1)	20 (15.2)	13 (9.8)
Anemia	1 (0.7)	0	2 (1.5)	1 (0.8)
Chronic obstructive pulmonary disease	1 (0.7)	1 (0.7)	0	0
Ecchymosis	1 (0.7)	0	0	0
Femoral neck fracture	1 (0.7)	0	0	0
Interstitial lung disease	1 (0.7)	0	1 (0.8)	1 (0.8)
Right ventricular failure	1 (0.7)	1 (0.7)	0	0
Thrombocytopenia	1 (0.7)	1 (0.7)	0	0
Back pain	0	0	1 (0.8)	0
COVID-19	0	0	1 (0.8)	0
Cardiorespiratory arrest	0	0	1 (0.8)	1 (0.8)
Erythema	0	0	1 (0.8)	0
Flank pain	0	0	1 (0.8)	0
Infection	0	0	1 (0.8)	1 (0.8)
Infusion-related reaction	0	0	1 (0.8)	1 (0.8)
Ischemic stroke	0	0	1 (0.8)	1 (0.8)
Myocardial ischemia	0	0	1 (0.8)	1 (0.8)
Neutropenia	0	0	4 (3.0)	4 (3.0)
Pneumonia	0	0	1 (0.8)	0
Rash erythematous	0	0	1 (0.8)	0
Sepsis	0	0	1 (0.8)	0
Skin toxicity	0	0	1 (0.8)	0
Subcutaneous abscess	0	0	1 (0.8)	0
Tumor lysis syndrome	0	0	1 (0.8)	1 (0.8)
Urosepsis	0	0	1 (0.8)	1 (0.8)
Vomiting	0	0	1 (0.8)	0

Abbreviations: AE, adverse event; BendaR, bendamustine plus rituximab; TEAE, treatment-emergent AE.

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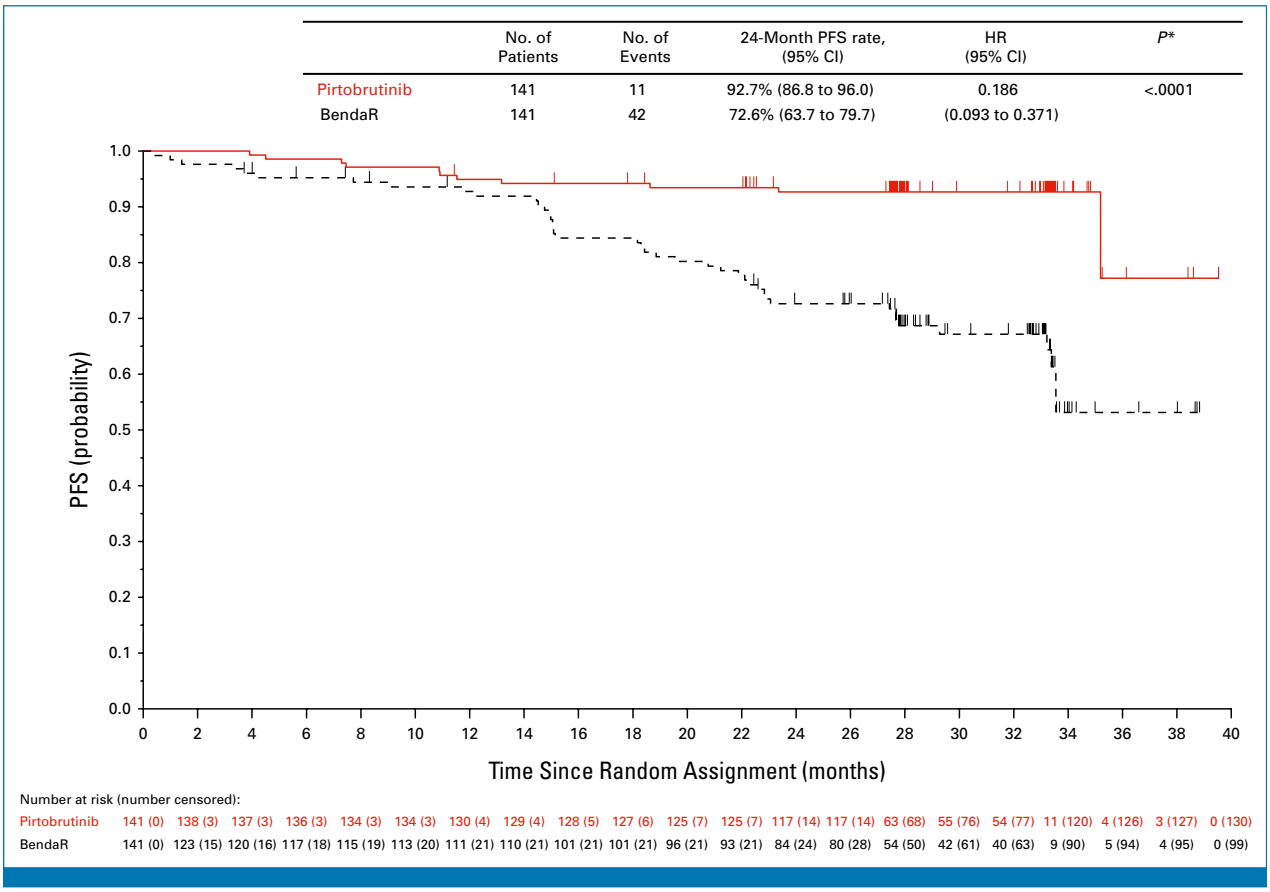

FIG A1. Kaplan-Meier curve of INV-assessed PFS. *Nominal *P* value. BendaR, bendamustine plus rituximab; HR, hazard ratio; INV, investigator; IRC, independent review committee; PFS, progression-free survival.

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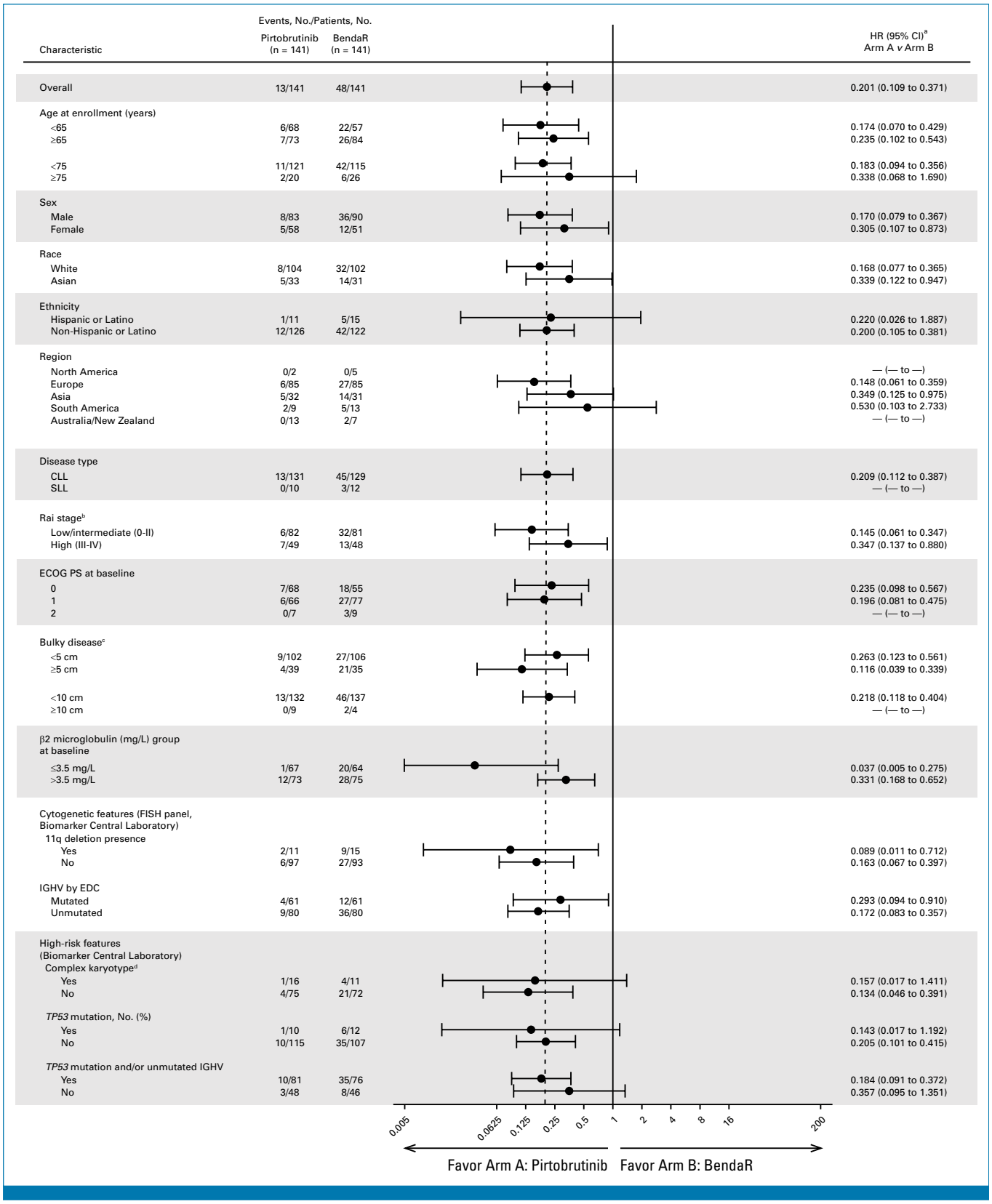

FIG A2. Forest plot of IRC-assessed PFS across patient subgroups. [a]HR and 95% CI based on the unstratified Cox regression model. [b]Rai stage is summarized for patients with CLL as reported in the EDC. [c]The <5 cm and <10 cm categories include patients with no measurable lymph node. [d]Complex karyotype was defined as ≥3 chromosomal abnormalities. BendaR, bendamustine plus rituximab; CLL, chronic lymphocytic leukemia; ECOG, Eastern Cooperative Oncology Group; EDC, electronic data capture; FISH fluorescence in situ hybridization; HR, hazard ratio; IGHV, immunoglobulin heavy chain gene; IRC, independent review committee; SLL, small lymphocytic lymphoma.

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